

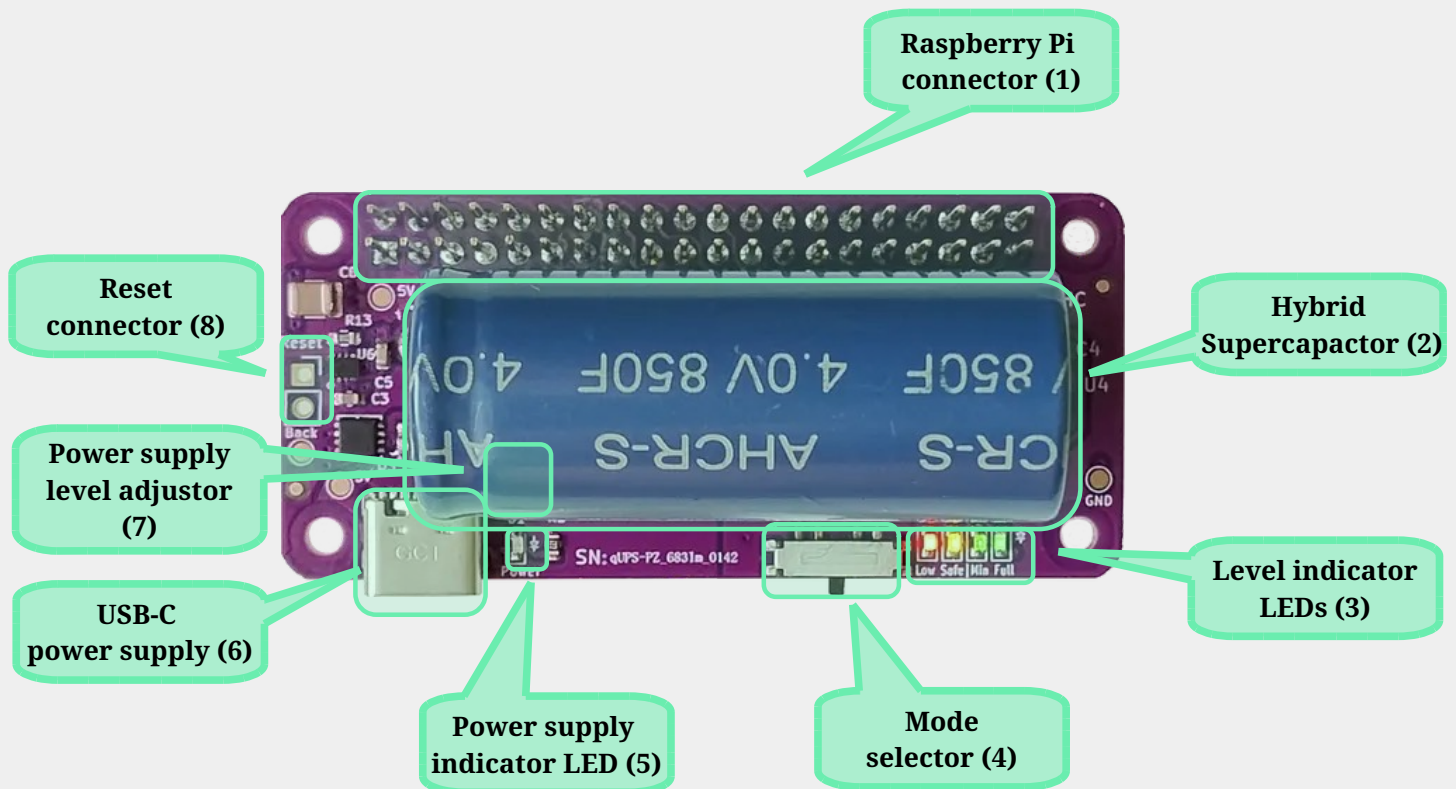


Stubborn Balance Zero v1.0

User Manual v1.0
5/2/2026

Please read the instruction manual for safe use and a user experience tailored to your needs.

The **Stubborn Balance Zero** has several connection, setup and feedback points for using it, highlighted in the figure below. For easy identification, the "()" indicates the subsequent reference numbers.



1 Safety regulations

1.1 Personal safety



The **Stubborn** product has an energy storage system that can be energised even when not connected to the mains.

The **Stubborn Balance Zero** has no replaceable parts other than the battery - only the manufacturer or an accredited service centre can carry out repairs and maintenance.

1.2 Product safety



The **Stubborn** product should be protected from too high or too low temperatures, direct sunlight. It should be kept in a dry place for 24 hours before installation.

Conductive liquids, plastic materials may cause short circuiting and permanent product damage, therefore avoid installation in such environments.

The **Stubborn Balance Zero** system should only be powered through the (6) connector.



POWERING THE PROTECTED DEVICE FROM ANOTHER SOURCE IS FORBIDDEN!!

Any **Stubborn** product must not be operated together with another **Stubborn** product and/or other uninterruptible power supply via connector (1) (40-pin terminal) !

1.3 Precautions

The system operates from 5V, which is low voltage, so it is protected from electric shock in a life-safety sense. In case of short circuits caused by foreign matter, the contacts may heat up and cause injury!

2 Introduction

Thank you for choosing AQEX **Stubborn Balance Zero** to protect your electronic device!

The **Stubborn** family has been carefully and thoroughly designed to provide the most efficient way to ensure smooth operation in a wide range of conditions, to meet a wide range of user requirements.

The **Stubborn** family is designed for uninterruptible operation of Raspberry Pi **Zero** compatible microcomputers. It can also protect any device requiring 5V DC with a maximum power consumption of 2.5 A. Hereafter, we will refer to the connected product collectively as "Protected Device".

The **Stubborn Balance Zero** is the hybrid supercapacitor version of the family. Its primary advantage is a maintenance-free, extreme-longevity power supply. For energy storage, it utilizes specialized **Hybrid Supercapacitors (HSC)**, which guarantee significantly higher charge cycles and superior thermal stability compared to traditional batteries. The uninterruptible backup time is determined by the selected capacity (450F, 850F, or 1100F) and the current draw of the connected device.

HSC capacity [F]	No load [min]	100% load [min]
450	46	13
850	90	22
1100	105	28

Table 1: Expected operating time [minutes]

2.1 Optimal circumstances of use

Thanks to the high capacity of the battery connected to the **Stubborn Balance Zero**, the product can be used in places where several hours of power outage or power surges can compromise reliable operation. With a sufficiently large battery capacity, several days of operation are guaranteed.

The **Stubborn** family is suitable for microcomputers, single board computers, microcontrollers and any 5V DC powered devices:

- for protection against power failure
- for overvoltage protection
- to achieve safe disconnection
- to ensure tasks and communication before disconnection
- for safe restart after a power failure

2.2 Non optimal circumstances of use

It is not suitable for protecting high-power computers due to the device's maximum current drain of 3.5A (17.5W power consumption). Other products are better suited for higher current needs.

3 Commissioning

The product is ready for use immediately after preparation and unpacking. For computers using a Raspberry PI **Zero** compatible 40 pin header, the connection is plug-and-play based, while in other cases the two pin +/- connection can be used.

3.1 Power supply

The **Stubborn Balance Zero** requires a 5V DC, min. 2A (min. 3A recommended) power supply. For 5V power supply, use a cable with USB-C (7) connector.

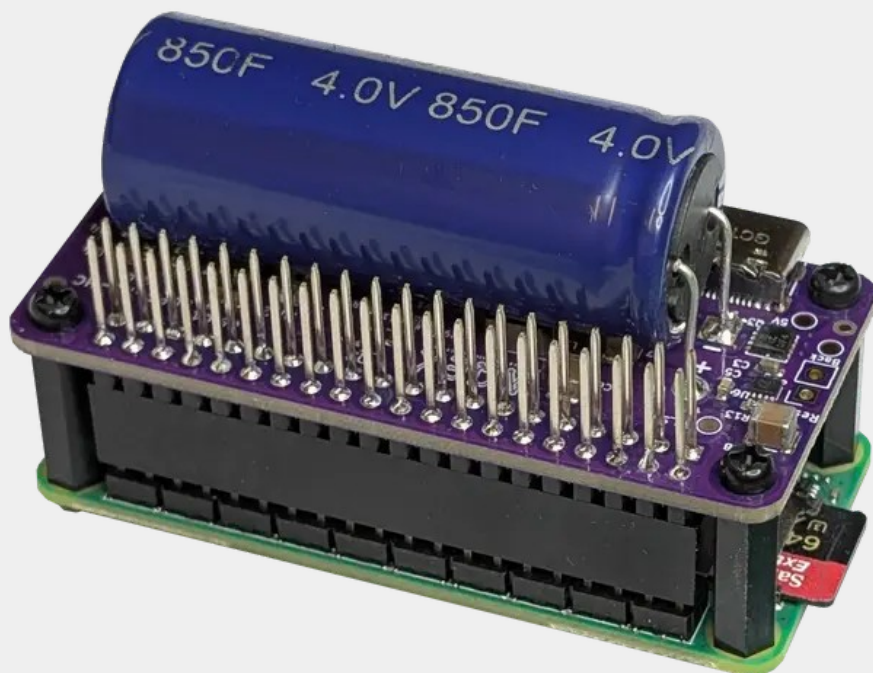
See Appendix for a list of tested power supplies and Troubleshooting.

3.2 Connections

The protected device can be connected to the product in different ways.

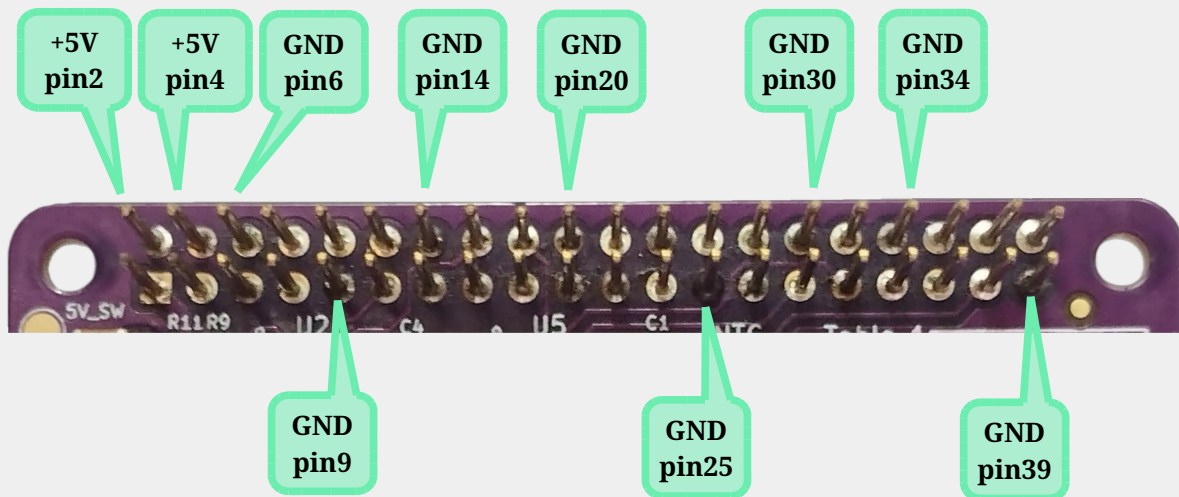
3.2.1 Single board computers (SBC)

If the SBC is equipped with a Raspberry PI (**Zero**) compatible 40 pin header, the **Stubborn Balance Zero** products are suitable for „Zero size” (pHAT) connection. Simply plug the product's (1) socket onto the computer's pin header as shown in the picture below.



3.2.2 Other devices

Any device that operates from 5 V DC can be connected to the product using the appropriate contacts on the 40-pin header (1).



3.2.3 Powering the unit

The **Stubborn Balance Zero** should be powered via USB-C connector (6), from the standard 5V – 5.2V power supply.

The presence of power is indicated by LED (5).



Safety power supplies that disable 5V at their output in the absence of a connected device (e.g. official Raspberry Pi 5 PSU) are suitable for powering the **Stubborn Balance Zero**.

3.2.4 GPIO pins

All GPIO pins of the protected device connected via the 40 pin row (1) can be accessed on the top side of the **Stubborn** family. With the exception of the three GPIO pins used by the **Stubborn Balance Zero**, all of the others are available for use for any other purpose.

3.2.5 Reset

The main function of the Reset connector's (8) is the momentary power interruption. It is possible to solder in a cable, push button, standard 2.54 mm pin header or switch. If a short circuit occurs between the two contacts, the **Stubborn Balance Zero** will cut the power and the operation of the protected device will be stopped for the duration of the short circuit (reset function).

The contact is polarity independent.



Voltage control is not allowed, only contact („short-circuit” or „open-circuit”) can be used.

Via the Reset connector (8), it is possible to control the power supply of the protected device from an external source, switching it off permanently is also allowed.

3.2.6 Communication

A **Stubborn Balance Zero** uses three GPIO pins for data exchange with the computer.

- Power Good - (pfo): Pin16 / GPIO 23
- Battery low voltage level – (lim): Pin18 / GPIO 24
- Shutdown - (shd): Pin 22 / GPIO 25

Pin functions:

- pfo: „power good”

When the voltage on the pin is high (logic 1, TRUE), the protected device is powered from the mains, and when it is low (logic 0, FALSE), the **Stubborn Balance Zero** product does not receive power from the mains, so it uses the energy stored in the battery to power the protected device.

- lim: „limit”, battery charge state is above the limit

If the voltage measured on the terminal is high (logic 1, TRUE), the energy stored in the battery is above the critical level. If the voltage level on the leg is low (logic 0, FALSE), the energy stored by the **Stubborn Balance Zero** has reached a critical level and a safe shutdown process should be initiated if required.

- shd: „shutdown”, protected device is operational

Indication to the **Stubborn Balance Zero** that the protected device is in operation. Its purpose is to ensure that the device restarts correctly after software shutdown, when power is restored and the energy storage has reached the correct charge level.

The protected device indicates to the **Stubborn Balance Zero** that it is operational by pulling up the "shd" pin (logic 1, TRUE). Loss of external power and the energy storage falling below critical level may result in software shutdown of the protected device. In this case, the pin will be set to a low voltage level (logic 0, FALSE). This will signal to the **Stubborn Balance Zero** that the protected device must be restarted in any case when external power is restored. If the energy storage is below the low charge level, the system will first recharge it to high charge level and then power on the protected device.

3.2.7 Hybrid supercapacitor charging

If the power supply cannot handle the combined load of the 2A charging current and the current drawn by the protected device, replace the power supply.



The external power supply may be able to operate the Raspberry Pi **Zero** without the **Stubborn Balance Zero**, but not through it. The reason can be that the power supply does not have enough power to meet the increased current demand caused by the battery charging.



It is recommended to avoid a lower quality power supply. When used, current consumption below the indicated maximum load may cause voltage stability problems. If the voltage drops below the level tolerated by the protected device, shutdown or unstable operation will occur.

3.3 Setup

3.3.1 Hardware setup

3.3.1.1 Potentiometer (7) for adjusting the threshold of the input voltage

The reason for this adjustability is to enable the device to operate optimally with different power supply type and any protected device pair.

The threshold voltage of the external power supply can be adjusted by the potentiometer (7). Below this voltage level, the product switches to uninterruptible mode and above it to external power supply.



If the level is set too high, the device will switch between modes unnecessarily, and in extreme cases it may switch permanently.



If the level is set too low, the device will not switch to uninterruptible mode in a timely manner and the Raspberry Pi **Zero** may shut down.

3.3.1.2 Mode switch (4)

The mode switch has three positions, which are indicated on the panel.

- „OFF” mode

The **Stubborn Balance Zero** will remove power from the protected device (regardless of the presence of an external power source and the power level of the energy storage device).

- „ON” mode

The **Stubborn Balance Zero** shall provide power to the protected device if an external power source is available or if there is sufficient power in the energy storage device.



If the charge level of the energy storage device is low and the external power supply is cut off, the protected device will not be able to perform a guaranteed regular load/unload cycle.

- „AUTO” (automatic) mode

The system shall guarantee the execution of a regular boot/shutdown cycle regardless of the loss of external power supply.

3.3.2 Software settings

The **Stubborn** family is able to share information with the protected device about the power supply and its own status. This allows the protected device to safely perform the necessary steps for saving and exit before the battery is completely discharged.

The exact details of the power-off and power-on function can be set in software.

The feedback pins in [chapter 3.2.8](#) are intended to provide information to the device and to perform adjustments on the **Stubborn Balance Zero**.

The control program can be downloaded from AQEX github:
<https://github.com/aqexhu/qups-guard>

A required parameter is the configuration for the GPIO pins used for communication, which for the **Stubborn Balance Zero** is:

```
qups-guard --dip 01
```

Optional parameter for delaying shutdown:

```
--shutdown-delay 20
```

More detailed and continuously updated description on the GitHub page.

4 Usage

4.1 Operation

If properly connected, the system will work under normal conditions with default settings.

The **Stubborn Balance Zero** will start charging the energy storage after the power supply is connected.

4.2 Operating characteristics and safety warnings

4.2.1 AUTO mode: Intelligent start protection

In AUTO mode, the primary concern of the **Stubborn Balance Zero** is to maintain system integrity. The protected device (Raspberry Pi **Zero**) is only powered when the battery charge has reached a safe level that provides sufficient energy for the software shutdown in the event of an immediate power failure.

What can the user expect?

- **Startup delay:** After connecting the power supply, the system does not start immediately. This waiting time can take up to several minutes, depending on the state of the deeply discharged battery, the capacity of the power supply, and internal settings.
- **Process:** The system first charges the battery to the critical threshold and only then allows the Raspberry Pi **Zero** to start up.

Advantage: This mode of operation ensures that the Raspberry Pi **Zero** does not enter a boot loop (when the voltage immediately drops below the critical level due to the start-up current consumption and the machine shuts down again).

4.2.2 Operation and risks of ON mode

In ON mode, the **Stubborn Balance Zero** immediately powers up and starts the connected Raspberry Pi **Zero**, regardless of the current battery charge level or the status of the external power supply.

Safety warning: In this mode, software protection features are limited. If the external power supply fails and the battery reaches a critical level before the system shuts down, the Raspberry Pi will shut down **immediately and unexpectedly**.

- **Consequence:** Sudden power loss may result in damage to the operating system (SD card) or data loss.
- **Recommendation:** Use ON mode only for testing purposes or with a stable power supply. AUTO mode is recommended for continuous operation.

4.2.3 Specific features of battery voltage measurement and shutdown process

The voltage measured at the battery terminals does not correspond to the actual chemical voltage of the cell due to internal resistance and system losses (wires, connectors). This difference increases proportionally with the increase in load current.

This phenomenon can be critical when shutting down the system:

- **Low voltage detection:** When the battery is low, the software reaches the threshold value and initiates a safe shutdown of the Raspberry Pi **Zero**.
- **Voltage rebound:** At the end of the shutdown process, the Pi's current consumption drops dramatically. Due to the removal of the load, the measured battery voltage rises suddenly.
- **The fault phenomenon:** If this increased voltage exceeds the shutdown threshold, the **Stubborn Balance Zero** controller detects that the battery has not actually run out, so it does not interrupt the power supply. As a result, the Raspberry Pi **Zero** remains in "shutdown" mode (it is powered but not running) and does not restart.

Solution: A shutdown delay can be set in the **Stubborn Balance Zero** software. This timer ensures that the system continues to draw power from the battery for a sufficient period of time after the shutdown command is issued, so that the voltage remains stable below the critical level even after the load is removed, thus ensuring complete shutdown.

4.2.4 Drip mode and restart threshold

The charging circuit uses trickle charging to protect the battery when the voltage drops below 2.8–2.9 V. In this state, the charging current is much lower than normal, which affects the system's readiness for operation.

Effect of the phenomenon: If the battery is deeply discharged, it takes **significantly longer** to reach the voltage level required for restarting, as the charging process starts with limited trickle charging.

4.2.5 Startup instability with a deeply discharged battery

When recharging a completely discharged battery, the system may have difficulty starting up, as indicated by the flashing "Low" (red) LED.

Cause of the phenomenon: This typically occurs when the input power supply is insufficient to handle the sudden inrush current.

1. The load causes the supply voltage to drop, which causes the input voltage sensor to disable charging for safety reasons.
2. When the load is removed, the power supply voltage is restored, the system attempts to charge again, and the process repeats.

This cycle usually stabilizes itself within **5–10 seconds**, once the system's internal buffers and the battery reach their minimum levels.

4.2.6 Important warning regarding storage

The **Stubborn Balance Zero**'s own control circuits and indicator LEDs consume minimal power from the battery even when switched OFF. This continuous load typically drains the battery within a few days, depending on the battery capacity and condition, without recharging.

Deep Discharge Protection: To ensure operational safety, the system disconnects the load at a 2.5V threshold (managed by the charging IC). It is important to note that **Hybrid Supercapacitor (HSC)** technology is not sensitive to deep discharge; it can withstand a 0V state without damage, allowing for immediate recommissioning even after long-term storage.

4.2.7 Switch between mains and battery power

The **Stubborn Balance Zero** detects the loss of external power supply or a voltage drop below the level set by [potentiometer \(7\)](#). In this case it automatically switches to battery power. When the external power supply is restored, it will switch back and recharge the energy consumed from the battery.



The switchover from external power to battery mode is extremely fast (**100-300 µs**). Although a momentary voltage drop may occur — triggering an '*Under-voltage detected!*' warning on the Raspberry Pi — it will not cause malfunctions or reboots due to the short duration.



Figure 1: Raspberry Pi Zero at 50% load

No alarm in the event of a switch back to mains power.

4.3 Intelligent functions

The **Stubborn Balance Zero** provides real-time power status and battery charge levels to the protected device, ensuring that during extended outages, the system receives a timely warning to initiate a safe shutdown before the batteries are depleted. Upon sensing the shutdown signal of the protected device, the **Stubborn Balance Zero** cuts the power to guarantee a clean reboot and prevent the system from becoming stuck in a halted state when external power returns.

4.3.1 Event handling

The operating system or program running on the protected device can be prepared to detect in real time the information communicated by the **Stubborn Balance Zero** and perform various tasks in this context.

Two types of status indications are continuously displayed on the **Stubborn Balance Zero**:

- presence of external power supply - pfo
- energy storage battery charge level - lim

Via the presetted GPIO pins, the service or program running on the device detects the changes and provides the possibility to trigger pre-set mechanisms.

For more information and utilities written in C and Python, see the <https://github.com/aqexhu/qups-guard> page.

4.4 Level indicator

The battery charge is indicated by the LED group numbered (3).

The five different coloured LEDs indicate the following battery power levels:

1. Low (red): low level, the device can be switched off at any time.



If the device is in "AUTO" mode and the corresponding SHD pin is at low level, this level will disconnect the power from the protected device when reached.

2. Safe (yellow): there is sufficient time for a safe shutdown.

3. Min (green): There is sufficient time for one start-up and one shutdown - in "AUTO" mode the protected device will start.

5. Max (green): Fully charged.



The LEDs indicate the above charge levels, not percentages. These are functional level indicators and do not reflect the battery charge linearly.

5 Appendix

5.1 Technical data

5.1.1 Energy storage Device

Energy storage	Hybrid Super capacitor (HSC)
Charge cycle	50.000
Operation temperature	-20 .. +65°C
Zero2HW idle backup time - 450F	46 minutes
Zero2HW full load backup time - 450F	13 minutes
Zero2HW idle backup time - 850F	90 minutes
Zero2HW full load backup time - 850F	22 minutes
Zero2HW idle backup time - 1100F	105 minutes
Zero2HW full load backup time - 1100F	28 minutes

5.1.2 Physical parameters

Form factor	Raspberry Pi HAT
Dimensions (450F)	65x56x21mm + 11 mm (az (1) csatlakozó)
Dimensions (850F)	65x56x24mm + 11 mm (az (1) csatlakozó)
Dimensions (1100F)	65x56x24mm + 11 mm (az (1) csatlakozó)
Stackable	
Replaceable battery	

5.1.3 Electric parameters

Charging current	2A
Maximum discharge current	7.5A
Maximum load current	3.5A
Output voltage	5V

5.1.4 Connectors

Main power input	USB-C
Auxillary power input	
Reset	

5.1.5 Power Supply

Main Input (USB-C)	5V
Auxiliary Input	
USB PD (RPI5) compatibility	

5.1.6 Communication

Method	3 GPIO port
GPIO port allocation	1 option
Disabling option	

5.1.7 Setup

Mode options	AUTO-BE-KI
Adjustable level threshold	
Adjustable input voltage threshold	

5.1.8 Control

External control of the power of Raspberry Pi Zero	
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5.1.9 Indication

Input Power	
Battery voltage level	4 LED

5.1.10 Software

Control software	C++ és Python
Open source	

5.2 *List of supported power adapters*

- RaspberryPi official power adapter [5V@2.4A](#)
- RaspberryPi 5 official power adapter 5V@5A
- Goobay 43651 5V@2.1A

5.3 Troubleshooting

Symptom	Cause of error	Solution
If an external power supply is present, the "External power LED" (6) is not lit	External power supply problem	Replace the power supply
Raspberry Pi Zero will not start if the "Min" LED is on	Mode switch (4) is on OFF state	Mode switch (4) must be set to ON or AUTO
The circuit does not charge the battery below 2.9V.	The circuit charges only with trickle charging.	See section 4.3.